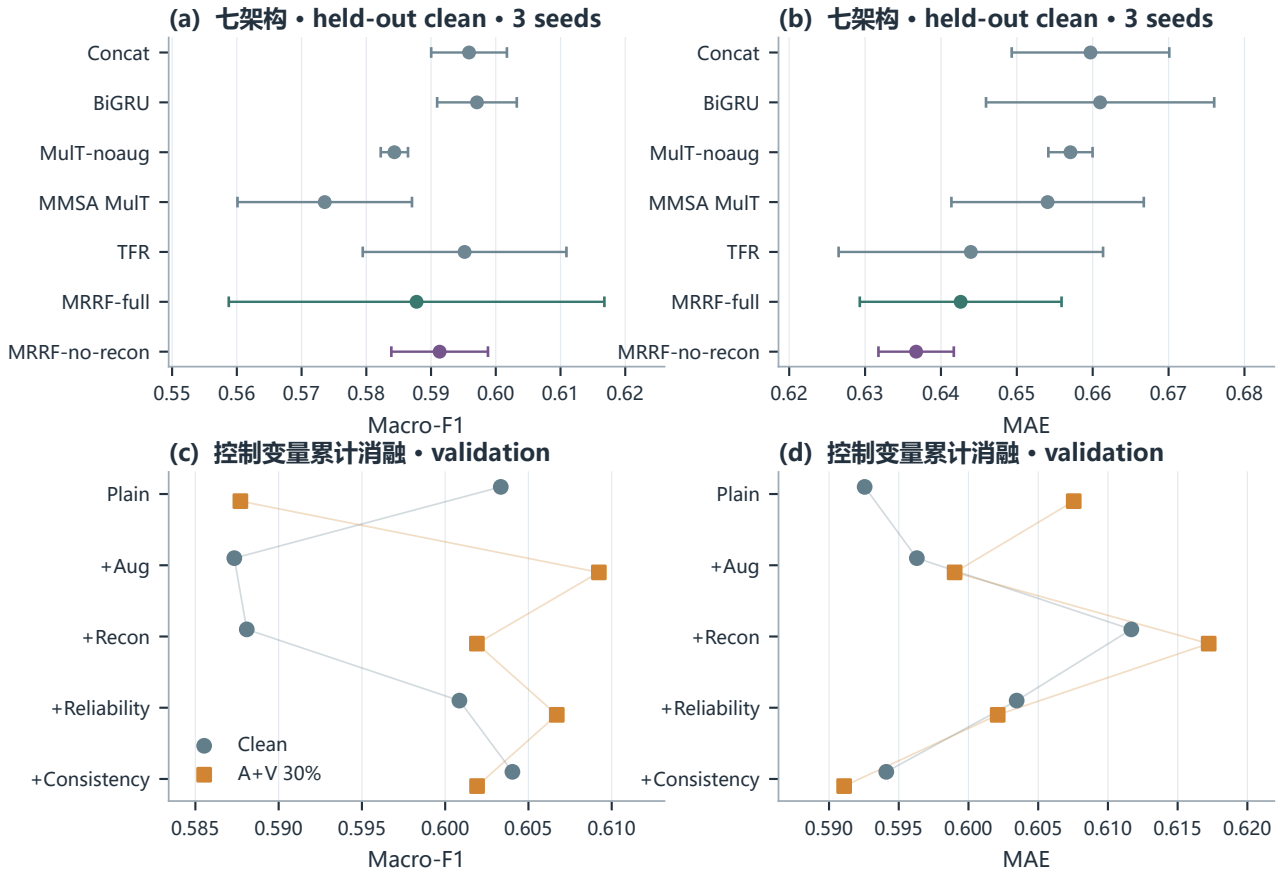
(a) 七架构 • held-out clean • 3 seeds
Concat
BiGRU
MulT-noaug
MMSA MulT
TFR
MRRF-full
MRRF-no-recon
0.55
0.56
0.57
0.58
0.59
0.60
0.61
0.62
Macro-F1
(b) 七架构 • held-out clean • 3 seeds
Concat
BiGRU
MulT-noaug
MMSA MulT
TFR
MRRF-full
MRRF-no-recon
0.62
0.63
0.64
0.65
0.66
0.67
0.68
MAE
(c) 控制变量累计消融 • validation
Plain
+Aug
+Recon
+Reliability
+Consistency
Clean
A+V 30%
0.585
0.590
0.595
0.600
0.605
0.610
Macro-F1
(d) 控制变量累计消融 • validation
Plain
+Aug
+Recon
+Reliability
+Consistency
0.590
0.595
0.600
0.605
0.610
0.615
0.620
MAE